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Section - 4 Grp - 1

Single Slit Diffraction

Date of experiment: **Partner's name:**

Date of Lab. report submission:

Aim of the Experiment

To study the intensity distribution of Fraunhofer diffraction pattern by a single slit and measure the width of the slit for a given wavelength of laser light.

Apparatus Required

(i) Optical Bench.
(ii) Laser Light Source (He-Ne 1mW).
(iii) Focusing lenses and adjustable slit.
(iv) Photometer (Optical base plate) & Screen
(v) Digital Multi-meter with connecting wires.

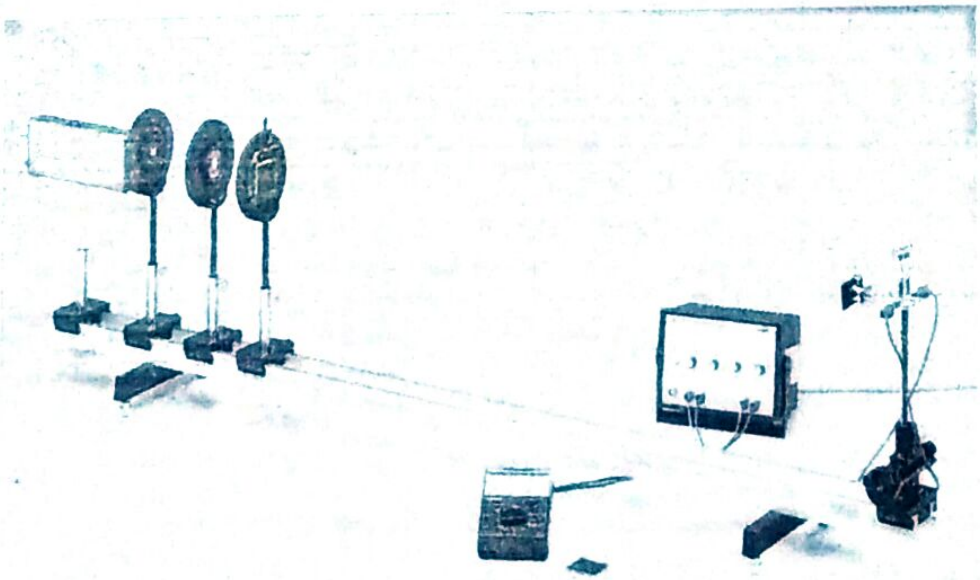


Fig-1: Experimental setup

(x) Plot intensity distribution of the diffraction as a function of θ parallel to the plane of the slit. Find the width of given slit from the plot and compare this width with that measured by microscope.

Observations

Vernier constant of micrometer= 0.01 mm

Wave length of the laser light = 632.8 nm

Distance of the slit from the photometer = 94 cm

Table-1: Intensity distribution of single slit pattern(x_0 corresponds to position of central maximum)

Sl. No.	Position (x cm)	$\theta_{lr} = (x-x_0)/D$ $x_0 = 17.25$ mm	Photo current/voltage (mA/V) [Left to right]	Position (x cm)	$\theta_{rl} = (x-x_0)/D$	Photo current/voltage (mA/V) [Right to left]
1	0	-0.1835	0.69	0	-0.1835	0.47
2	0.5	-0.1782	0.85	0.5	-0.1782	0.59
3	1	-0.1729	0.9	1	-0.1729	0.92
4	1.5	-0.1676	1.38	1.5	-0.1676	1.33
5	2	-0.1623	1.89	2	-0.1623	1.92
6	2.5	-0.1570	2.31	2.5	-0.1570	2.32
7	3	-0.1516	2.7	3	-0.1516	2.56
8	3.5	-0.1463	2.89	3.5	-0.1463	2.53
9	4	-0.1410	2.22	4	-0.1410	2.25
10	4.5	-0.1357	1.85	4.5	-0.1357	1.72
11	5	-0.1303	1.38	5	-0.1303	1.09

Sl. No.	Position (x cm)	$\theta_{lr} = (x-x_0)/D$	Photo current/voltage (mA/V) [Left to right]	Position (x cm)	$\theta_{rl} = (x-x_0)/D$	Photo current/voltage (mA/V) [Right to left]
12	5.5	-0.1250	0.81	5.5	-0.1250	0.72
13	6	-0.1197	0.56	6	-0.1197	0.57
14	6.5	-0.1144	0.82	6.5	-0.1144	0.85
15	7	-0.1090	1.4	7	-0.1090	1.79
16	7.5	-0.1037	2.88	7.5	-0.1034	3.11
17	8	-0.0984	4.37	8	-0.0984	4.67
18	8.5	-0.0931	5.97	8.5	-0.0931	5.79
19	9	-0.0878	7.1	9	-0.0878	6.79
20	9.5	-0.0829	7.37	9.5	-0.0829	5.84
21	10	-0.0771	6.66	10	-0.0771	4.41
22	10.5	-0.0718	5.22	10.5	-0.0718	2.37
23	11	-0.0665	3.79	11	-0.0665	1.38
24	11.5	-0.0612	2.26	11.5	-0.0612	2.07
25	12	-0.0558	2.2	12	-0.0558	5.57
26	12.5	-0.0505	4.5	12.5	-0.0505	13.57
27	13	-0.0452	10.71	13	-0.0452	25.6
28	13.5	-0.0399	22.71	13.5	-0.0399	41.6
29	14.	-0.0346	37	14	-0.0346	62.2
30	14.5	-0.0293	53.9	14.5	-0.0293	86.4

Sl. No.	Position (x cm)	$\theta_{lr} = (x-x_0)/D$	Photo current/voltage (mA/V) [Left to right]	Position (x cm)	$\theta_{rl} = (x-x_0)/D$	Photo current/voltage (mA/V) [Right to left]
31	15	-0.024	73.9	15	-0.0240	108.6
32	15.5	-0.0187	99.3	15.5	-0.0187	115.8
33	16	-0.0133	115.2	16	-0.0133	116
34	16.5	-0.0080	117.2	16.5	-0.0080	116.9
35	17	-0.0027	118.2	17	-0.0027	117
36	17.5	0.0027	118.2	17.5	0.0027	117
37	18	0.0080	117.8	18	0.0080	110.4
38	18.5	0.0133	117.6	18.5	0.0133	103
39	19	0.0187	115.3	19	0.0187	99
40	19.5	0.0240	100.37	19.5	0.0240	88.1
41	20	0.0293	84.5	20	0.0293	80.41
42	20.5	0.0346	65.5	20.5	0.0346	66.2
43	21	0.0399	45.1	21	0.0399	51.1
44	21.5	0.0452	28.58	21.5	0.0452	36.17
45	22	0.0505	15.45	22	0.0505	23.62
46	22.5	0.0558	7.53	22.5	0.0558	14.03
47	23	0.0612	2.66	23	0.0612	6.47
48	23.5	0.0665	0.93	23.5	0.0665	2.48
49	24	0.0718	1.25	24	0.0718	0.72

Sl. No.	Position (x cm)	$\theta_{lr} = (x-x_0)/D$	Photo current/voltage (mA/V) [Left to right]	Position (x cm)	$\theta_{rl} = (x-x_0)/D$	Photo current/voltage (mA/V) [Right to left]
50	24.5	0.0771	2.94	24.5	0.0771	0.79
51	25	0.0825	4.59	25	0.0825	1.77
52	25.5	0.0878	5.87	25.5	0.0878	3.13
53	26	0.0931	6.34	26	0.0931	4.2
54	26.5	0.0984	6.24	26.5	0.0984	4.84
55	27	0.1037	6.16	27	0.1037	4.76
56	27.5	0.1090	5.46	27.5	0.1090	4.09
57	28	0.1144	3.86	28	0.1144	3.18
58	28.5	0.1197	2.64	28.5	0.1197	2.14
59	29	0.1250	1.66	29	0.1250	1.35
60	29.5	0.1303	0.73	29.5	0.1303	0.76
61	30	0.1357	0.52	30	0.1357	0.5
62	30.5	0.1410	0.76	30.5	0.1410	0.57
63	31	0.1463	1.15	31	0.1463	0.84
64	31.5	0.1516	1.68	31.5	0.1516	1.33
65	32	0.1570	2.06	32	0.1570	1.6
66	32.5	0.1623	2.22	32.5	0.1623	1.75
67	33	0.1676	2.18	33	0.1676	2.15
68	33.5	0.1729	2.88	33.5	0.1729	1.98

69	34	0.1782	1.35	34	0.1782	1.66
70	34.5	0.1835	0.96	34.5	0.1835	1.13
71	35	0.1888	0.64	35	0.1888	0.82
72	35.5	0.1941	0.51	35.5	0.1941	0.57

Conclusions

Conclusion should incorporate

(xiii) Interesting results you found in the experiment.

(xiv) Lacunas /difficulties that you encountered.

(xv) Practical applications/implications of the experiment.

(xvi) Any other issues.

Precautions

(i) One should not look directly to LASER light because it may damage the eye.

(ii) The optical bench should be levelled properly.

References

1. Fundamentals of optics by F. Jenkins and H. White.
2. Optics by A. Ghatak.
3. Introduction to Optics by E. Hecht
4. Feynman Lecturers.

Considerations

(i) What is diffraction of light?

(ii) How diffraction is different from interference?

(iii) Name different types of diffraction and their properties.

(iv) What will happen to the intensity distribution if the width of the slit is increased or decreased?

(v) What is the difference between single slit and double slit diffraction pattern?

Calculation

In case of Fraunhoffer diffraction, we have

slid width $b = \frac{\lambda \times 2D}{x_m}$

where λ is wavelength of light, D is distance of slit from the photometer

and x_m is distance between 2 minimas.

$\lambda = 632.8$ nm $D = 94$ cm

For Left to Right reading;

S.no.	x_m (cm)	$b_i = \frac{2D\lambda}{x_m}$	$(d_i = (\bar{b} - b_i)) \times 10^4$	$(d_i^2) \times 10^{-12}$
1	6	1.98×10^{-4}	-0.02	4
2	5.25	2.26×10^{-4}	0.26	676
3	6.25	1.90×10^{-4}	-0.1	100
4	6.5	1.89×10^{-4}	-0.17	289

From Right to Left reading,

5	5	2.38×10^{-4}	0.38	1444
6	6.25	1.90×10^{-4}	-0.1	100
7	6.75	1.76×10^{-4}	-0.24	576
8	6	1.98×10^{-4}	-0.02	4

$\bar{b} = 2\times10^{-4}$

$\sum d_i^2 = 3193\times10^{-12}$

$$s^2 = \frac{\sum d_i^2}{n} = \frac{3193\times10^{-12}}{72}$$

$$= 3.99\times10^{-12}$$

$$\sigma_m = \sqrt{\frac{s^2}{n-1}} = 7.55\times10^{-6}$$

$$\therefore\ b = \bar{b} \pm \sigma_m = (2\times10^{-4} \pm 7.55\times10^{-6})\,m$$

CONCLUSION:

1.) Value obtained for slit width is $(2\times10^{-4} \pm 7.55\times10^{-6})\,m$

2.) While performing the experiment, great care should be taken in focussing the source into slit.

3.) Diffraction and Interference property of light is shown.

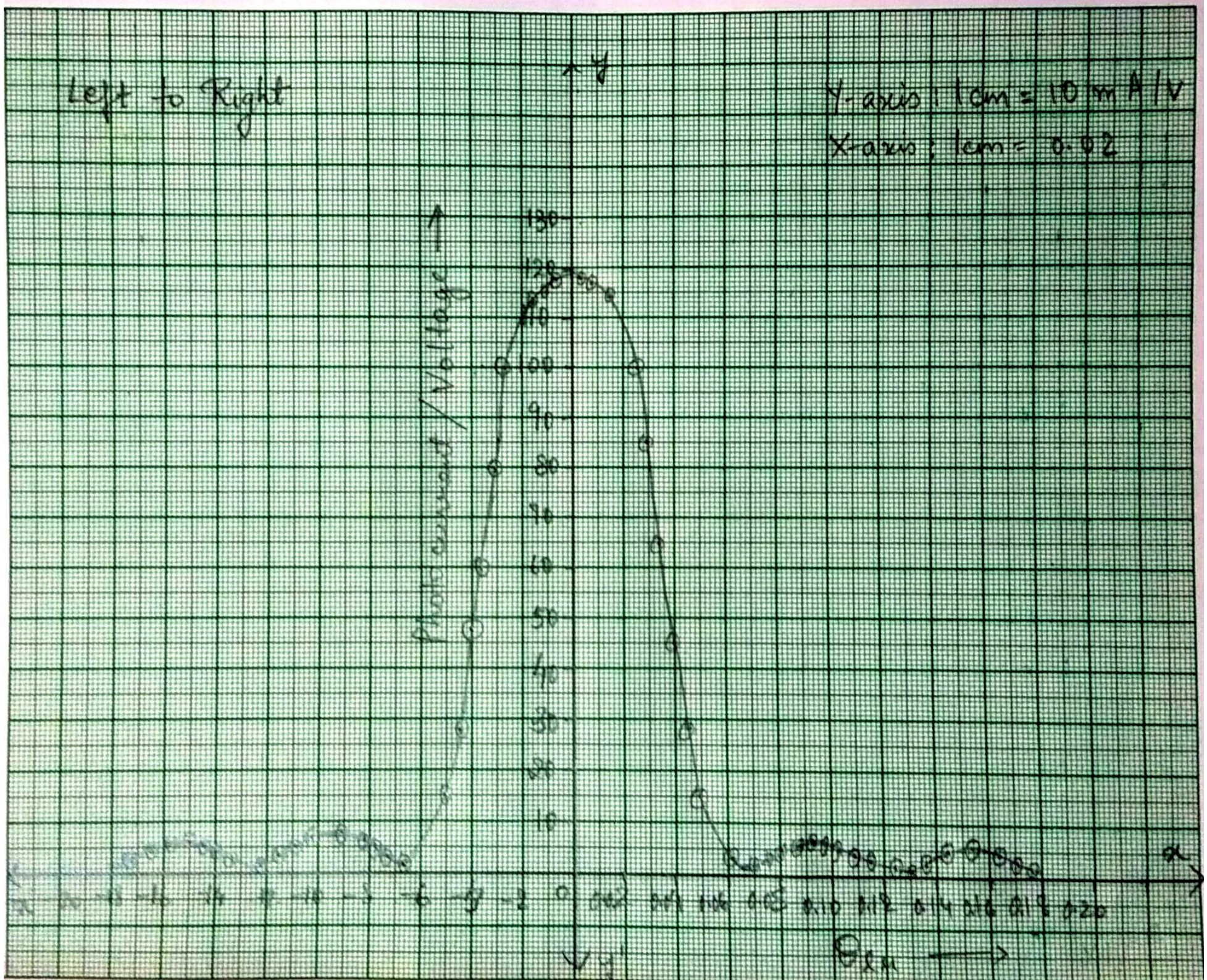
4.) It can also be modified to obtain the wavelength of light waves.

5.) One must not look directly at laser source as it may cause retina damage.

Left to Right

Y-axis: 1cm = 10 mA/V

X-axis: 1cm = 0.02



Right to left
Photo current / Voltage → (mA/V)
Scale of 10 units
Scale
X-axis : 1cm = 0.02 V
Y-axis : 1cm = 0.02 mA